TRANSNET
freight rail

A division of Transnet limited

ENGINEERING AND TECHNOLOGY
TECHNOLOGY MANAGEMENT

SPECIFICATION

Alarm limits for the measurements of weighbridges, wheel-impact monitors and skew bogie detectors

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CONTENTS

1 SCOPE

1.1 Identification

This document specifies the alarm limits for the different types of alarms available in the Integrated Train Condition Monitoring System (ITCMS) for measurements obtained from Weighbridges, Wheel Impact Monitors and Skew Bogie Detectors. This document also provides information regarding the severity of the detected defect and requirements for the operating actions.

1.2 Overview

Transnet is installing various train condition monitoring systems on its railway lines which are used to improve the safety of operations. This is done by generating alarms from either the condition monitoring system itself or by the ITCMS. The ITCMS has the ability to generate alarms by using the measurements of these systems and making certain calculations to determine the condition value. Some of the alarms are routed to an Operator Alarm Terminal (OAT) situated on the relevant Train Control Officer's (TCO) desk and some of the alarms are routed to a Maintenance Alarm Terminal (MAT) which is used for the scheduling of maintenance on the rolling stock.

In order to generate meaningful and valid alarms, the alarms limits need to be configured. Since there are different documents which specifies different alarms limits, this document will serve as the standard in Transnet and will specify the different alarm limits to be configured for these systems.

1.3 Document Overview

This document will serve as the standard for the specified alarm limits and will replace all other documents and specifications which specify an alarm limit with regards to the measurements of weighbridges, wheel impact monitors and skew bogie detectors installed on the railway network in Transnet.

All the different types of alarms will be defined and specified. These alarms are applicable on weighbridges, wheel impact monitors and skew bogie detectors. Appendix B contains a matrix with all the different alarms and their alarm limits.

1.4 Applicable documents

This document is based on the following documents:

1.4.1 BBB8790 (Version 2XA) – “ITCMS Measurements and alarms”

1.4.2 BBC7837 – “Vehicle/Track interaction – Force limits”

1.4.3 BBD7709 – “Limits for unbalanced loading of freight wagons”

1.4.4 BBD7313 – “Skidded wheels – skid length and wheel impact limits”

1.4.5 Ore line directives

These documents were used for literature purposes and for guidance to decide on the alarm limits. In addition to the information of these documents, a workgroup was constituted to verify and update the alarm limits specified in this document. The workgroup represented the following departments in Transnet Freight Rail:

- Technology Management -Track Technology
- Technology Management –Train Design
- Technology Management – Mechanical Technology
- Technology Management – Wheel-Rail Interaction
- Technology Management –Condition Assessment Technologies
- Infrastructure Engineering

This document will henceforth replace all other documents regarding the alarm limit values for weighbridges, wheel impact monitors and skew bogie detectors.

2 TYPES OF ALARMS SPECIFIED

The alarms-limits will be specified under different types of alarms. The following types of alarms are distinguished:

2.1 Type 1 Alarm (Continue to maintenance depot)

This alarm will be generated for non-critical operational conditions. Under these conditions it is understood that the situation is not safety-critical at the time (i.e. there is no danger of catastrophic incidents such as a derailment). These alarms require no operational actions and are presented to the maintenance department for action.

2.2 Type 2 alarm (Continue to station)

This alarm will be generated for serious conditions and requires operating intervention. These conditions are not safety-critical at the time, but have the potential to develop into a catastrophic condition if not attended to. Therefore, the train can continue to the next station for corrective actions before the train may depart.

2.3 Type 3 alarm (Stop Train Immediately)

This alarm is for safety-critical conditions and requires operating intervention. This means that the train shall be stopped immediately for corrective actions before the train can depart.

3 ALARM LIMITS

3.1 Skew loading (end-to-end)

3.1.1 Skew loading (end-to-end) is defined as uneven loading in the longitudinal direction, i.e. the difference between the mass of the front and back bogie of a wagon.

3.1.2 The ratio of end-to-end differences is calculated as follows:

$$\% \text{ end-to-end difference} = |LB1 - LB2|/(LB1+LB2) \times 100$$

where LB1 = Total load on front bogie
LB2 = Total load on back bogie

3.1.3 When the end-to-end difference is more than 12% a Type 2 alarm shall be generated. The 12% safety limit value includes the accuracy tolerance of the measuring system [BBD7709]. Commercial limits should be lower than 12%.

3.1.4 A deviation from the end-to-end skew loading limit as defined in 3.1.3 above is permitted for container wagons provided that the train compilation rules as defined in Report BBD7709 and Clause 1021.4 of the General Appendix No.6 (Part 1) are applied.

3.2 Skew loading (side-to-side)

3.2.1 Skew loading (side-to-side) is defined as uneven loading in the lateral direction, i.e. the difference between the mass of the left and right side of the wagon.

3.2.2 The ratio of side-to-side differences is calculated as follows:

$$\% \text{ side-to-side difference} = |LS1 - LS2|/(LS1+LS2) \times 100$$

where LS1 = Total load on left hand side wheels
LS2 = Total load on right hand side wheels

3.2.3 When the side-to-side difference is more than 12% a Type 2 alarm shall be generated. The 12% safety limit value includes the accuracy tolerance of the measuring system [BBD7709]. Commercial limits should be lower than 12%.

3.3 Overloading

3.3.1 Overloading is divided into two categories: Wagon overloading and train overloading.

- Wagon overloading is defined as the positive difference between the wagon's measured gross mass and the wagon's maximum allowable gross mass.
- Train overloading is defined as the positive difference between the measured gross mass of all the wagons of the train consist and the sum of all the wagons' maximum allowable gross mass, measured in tons.

3.3.2 When a wagon is overloaded by more than 7% of the maximum allowable gross wagon mass, a Type 2 alarm shall be generated. The 7% safety limit value includes the accuracy tolerance of the measuring system [BBD7709]. Commercial limits should be lower than 7%.

3.3.3 When the train is overloaded by more than n tons, where n is the number of wagons in the consist, a Type 2 alarm shall be generated.

3.4 Under-loading

3.4.1 Under-loading is relevant when an under-loaded wagon is in front of fully loaded wagon(s) in a train consist.

3.4.2 Under-loading is the positive difference between the maximum allowable gross wagon mass and the measured gross wagon mass, measured in tons.

3.4.3 When a wagon is under-loaded by more than 30% of its capacity, a Type 2 alarm shall be generated.

3.5 Wheel Impacts

Due to the fact that WIM-WIMs are installed on different class lines with different rail types which are able to withstand different wheel impact forces, the maximum wheel impact shall also be specified according to the class of line and the type of rail on which the WIM-WIM is installed.

With reference to the above and Report BBD7313, maximum wheel impacts as measured by a WIM-WIM system shall not exceed the limit values as provided in the table below:

Class of line	Rail section	Rail steel with lowest strength	Max. permissible axle load	Type 1 alarm	Type 2 alarm	Type 3 alarm
S	60 kg/m	Spoornet S116 CrMn	30 ton	-	240 kN (or 24 tons)	270 kN (or 27 tons)
N1	57 kg/m	UIC Grade 900A	22 ton	-	160 kN (or 16 tons)	190 kN (or 19 tons)

Note: For wheel impacts, a Type 3 alarm implies that the train shall stop, the brakes of the suspect vehicle shall be isolated, and then the train may travel to the nearest loop or station at a maximum speed of 30km/h. Here the required maintenance shall be conducted before the suspect vehicle can continue.

3.6 Lateral Forces

The different types of lateral force measurements are explained by means of the drawing in Figure 1.

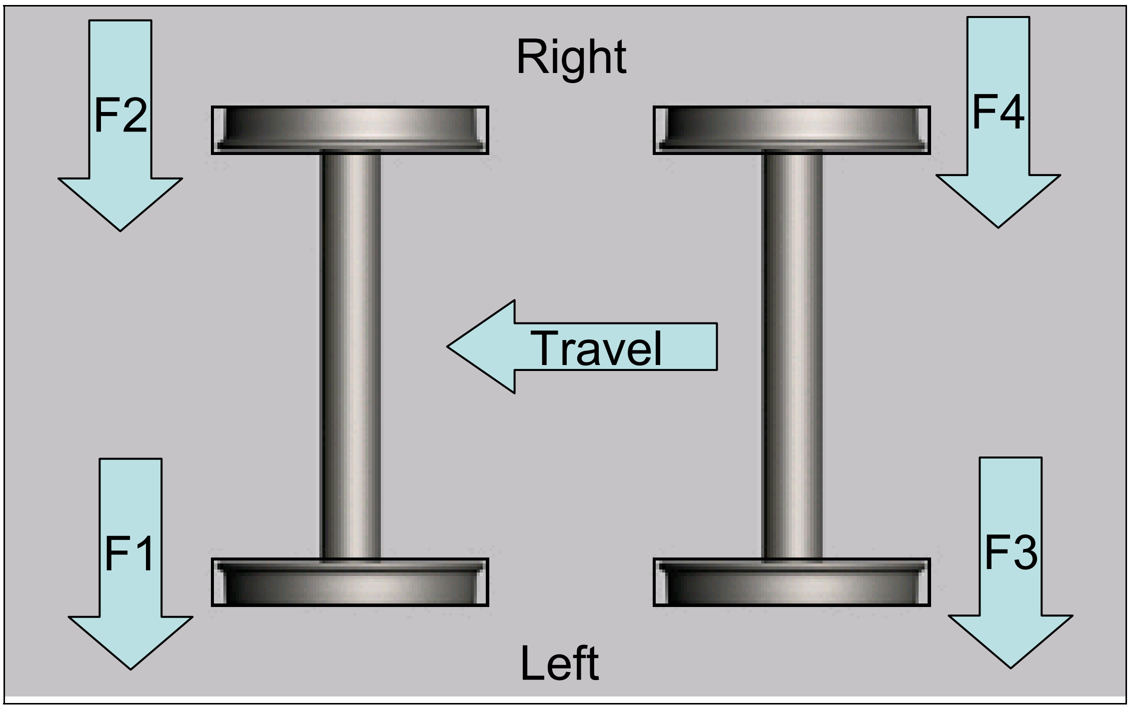

Figure 1: Lateral force convention (forces applied on the rail by the wheel)

According to Figure 1, the following is defined:

Wheel 1 lateral force = F1
Wheel 2 lateral force = F2
Wheel 3 lateral force = F3
Wheel 4 lateral force = F4

Axle 1 (wheel 1 & 2) gauge spreading force = F1 - F2
Axle 2 (wheel 3 & 4) gauge spreading force = F3 – F4
Bogie couple = (F1 + F2) – (F3 + F4)

For 3 axle bogies, bogie couple is calculated using the lateral forces from axle 1 and 3 (the outermost axles on the bogie).

3.6.1 When a wheel lateral force greater than 3 tons and less than 4 tons is detected, a Type 1 alarm shall be generated.

3.6.2 When a wheel lateral force greater than 4 tons and less than 6 tons is detected, a Type 2 alarm shall be generated.

3.6.3 When a wheel lateral force greater or equal than 6 tons or a Lateral / Vertical wheel force greater than 1 is detected, a Type 3 alarm shall be generated.

3.6.4 When a gauge spreading force greater 4 tons and less than 5 tons is detected, a Type 1 alarm shall be generated.

3.6.5 When a gauge spreading force of greater or equal than 5 tons is detected, a Type 2 alarm shall be generated.

3.6.6 When a bogie couple limit greater than 4 tons and less than 5 tons is detected, a Type 2 alarm shall be generated.

3.6.7 When a bogie couple limit greater or equal than 5 tons is detected, a Type 2 alarm shall be generated.

4 APPENDIX A

To follow is a glossary of terms which can assist in understanding the document better:

GLOSSARY OF TERMS

TERM	DEFINITION
ITCMS	Integrated Condition Monitoring System
MAT	Maintenance Alarm Terminal
OAT	Operator Alarm Terminal
TCO	Train Control Officer
Station	The closest place with a siding which will facilitate the uncoupling of wagons
Measured Gross Wagon Mass	The gross wagon mass as measured by the weighing instrument
Maximum Allowable Gross Wagon Mass	The maximum carrying capacity of a wagon + the wagon’s tare mass
Gross wagon mass	The tare mass of wagon + the mass of its load
Tare mass	The mass of the wagon when empty

5 APPENDIX B

Safety Alarms limits for rolling stock			
	ALARMS		
ALARM TYPE	**TYPE 1 (Continue to maintenance depot)**	**TYPE 2 (Continue to station)**	**TYPE 3 (Stop train)**
SKEW LOADING (END-END)		≥ 12%	
SKEW LOADING (SIDE-SIDE)		≥ 12%	
OVERLOADING		≥ 7% per wagon 1 ton x number of wagons on train	
UNDERLOADING		More than 30% under load, then at next depot remove wagon to back of train	
WHEEL IMPACTS		240 kN (or 24 tons) on a S 60 kg/m line 160 kN (or 19 tons) on N1 57 kg/m line	270 kN (or 27 tons) on a S 60 kg/m line 190 kN (or 19 tons) on N1 57 kg/m line
LATERAL FORCES	> 3 ton and < 4 and ton	> 4 and < 6 ton	≥ 6 tons or L/V > 1.0 Remove from train
GAUGE SPREADING FORCES	> 4 ton and < 5 ton	≥ 5 ton	
BOGIE COUPLE LIMITS	> 4 ton and < 5 ton	≥ 5 ton	

Table 2: Alarm Matrix